



On the HTC Vive, Valve's collaboration headset with HTC, you can see the positioning of the photoreceptors that use the Lighthouse tech.

relative to the light-emitting base tower. Needless to say, using multiple of these photosensors can read a 3D pose or orientation as well.

A large section of the VR fraternity believes that light field technology is also an answer. Because there is another problem to be dealt with as well: an eye focussing problem called vergence-accommodation conflict. It is something that big VR makers like Oculus and Sony are aware of, and are trying to find a solution to. So what's it all about? When you try to look at an object, firstly you 'Look', i.e point your iris in the direction of the object. If the object is close your irises converge, and diverge if they are far, hence 'vergence' is the term used. Secondly, you focus on the object, i.e accommodate the clarity of the object based on its distance. Normally, both vergence and accommodation happen in conjunction, since they are both trying to process the same object. The harmony is broken when you put your eyes into a VR headset. Usually, in a VR headset, the respective screens for each eye show an image that is slightly offset from the other to make it appear real - the higher the offset, the closer the object is. So your eyes are trying to converge on the object being shown. But, the screen is very close to your face, hence your eyes are trying to accommodate for that short distance and focussing on the entire image. This forceful uncoupling